

Titanic Survival Prediction – Task 3

Objective

The objective of this project is to predict whether a passenger survived the Titanic disaster using machine learning techniques and passenger information such as age, gender, class, fare, and family details.

Tools and Technologies Used

- Python
- Kaggle Notebook
- Pandas
- NumPy
- Seaborn
- Scikit-learn
- Matplotlib

Workflow

1. Imported required Python libraries
2. Loaded the Titanic dataset
3. Performed exploratory data analysis (EDA)
4. Checked missing values and dataset information
5. Visualized survival patterns using graphs and charts
6. Cleaned and preprocessed the dataset
7. Encoded categorical columns like gender and embarked port
8. Split the dataset into training and testing sets
9. Trained multiple machine learning models:
 - Logistic Regression
 - Decision Tree Classifier
 - Random Forest Classifier
 - Support Vector Machine (SVM)
 - K-Nearest Neighbors (KNN)
10. Compared model accuracies
11. Evaluated model performance using accuracy score and classification report
12. Saved the trained machine learning model file

Result

The machine learning models successfully predicted Titanic passenger survival with good accuracy, and the best-performing model produced reliable classification results.

Conclusion

This project demonstrated the complete machine learning workflow including data preprocessing, visualization, model training, evaluation, and prediction using the Titanic dataset.

Links

GitHub Repository Link:

<https://github.com/vasu-z/task-3-titanic-survival-prediction>

Notebook Link:

<https://www.kaggle.com/code/champ1z1/task-3-titanic-survival-prediction>

Model File Link:

<https://drive.google.com/file/d/1rlGp7Op8cxGN9CmH02u9IMvJsqKXVkWx/view>

Project Screenshots

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

import joblib

# Load Titanic Dataset
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"

df = pd.read_csv(url)

# Show Dataset
print("First 5 Rows:")
display(df.head())

# Dataset Info
print("\nDataset Information:")
df.info()

# Statistical Summary
print("\nDataset Description:")
display(df.describe())

# Missing Values
print("\nMissing Values:")
print(df.isnull().sum())
```

```
# Fill Missing Values
df['Age'] = df['Age'].fillna(df['Age'].mean())
df['Fare'] = df['Fare'].fillna(df['Fare'].mean())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

# Convert Categorical Data
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})
df['Embarked'] = df['Embarked'].map({'S': 0, 'C': 1, 'Q': 2})

# Drop Unnecessary Columns
df.drop(['PassengerId', 'Name', 'Ticket', 'Cabin'], axis=1, inplace=True)

# Show Cleaned Data
print("\nCleaned Dataset:")
display(df.head())

# =====
# DATA VISUALIZATION
# =====

# Survival Count
plt.figure(figsize=(6,5))
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()

# Age Distribution
plt.figure(figsize=(8,5))
sns.histplot(df['Age'], bins=20, kde=True)
plt.title("Age Distribution")
plt.show()
```

```
# Fare Distribution
plt.figure(figsize=(8,5))
sns.histplot(df['Fare'], bins=20, kde=True)
plt.title("Fare Distribution")
plt.show()

# Correlation Heatmap
plt.figure(figsize=(10,7))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

```
# =====
# MODEL TRAINING
# =====

X = df.drop('Survived', axis=1)
y = df['Survived']

x_train, x_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Logistic Regression Model
model = LogisticRegression(max_iter=1000)

model.fit(x_train, y_train)

# Predictions
y_pred = model.predict(x_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nModel Accuracy:")
print(f'{accuracy * 100:.2f}%')

# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# Save Model
joblib.dump(model, "titanic_model.pkl")

print("\nModel saved as titanic_model.pkl")
```

First 5 Rows:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 12 columns):
```

```
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Dataset Description:

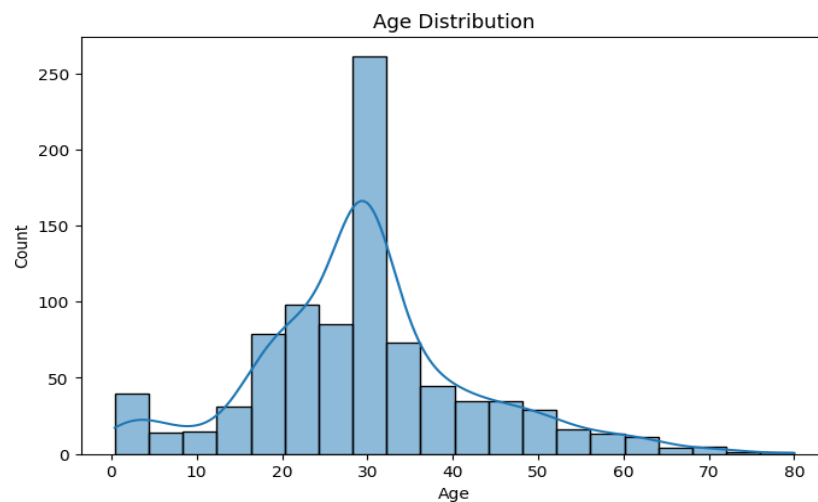
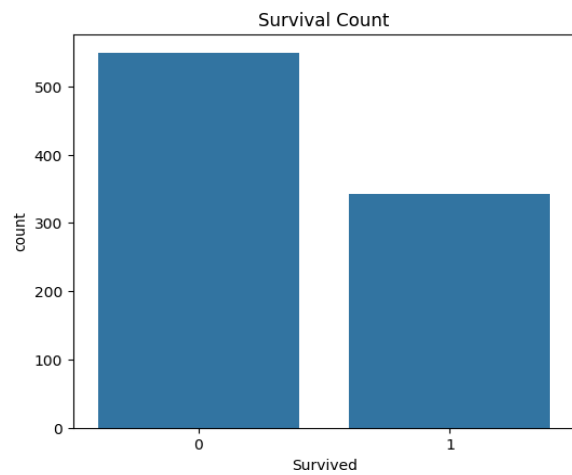
	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

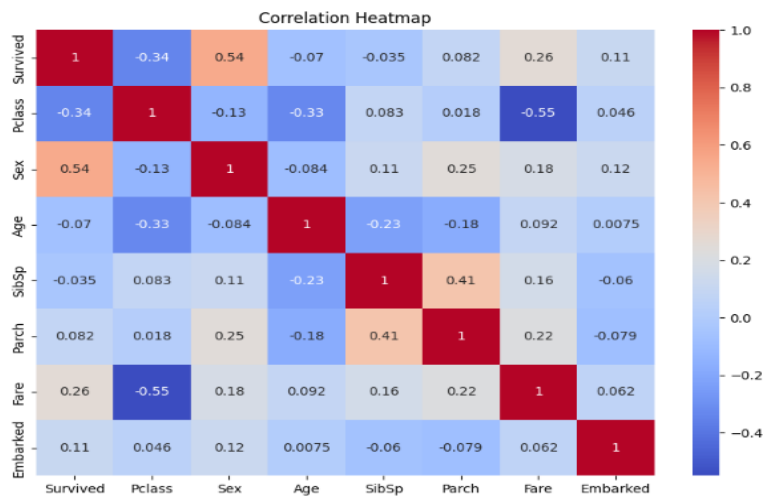
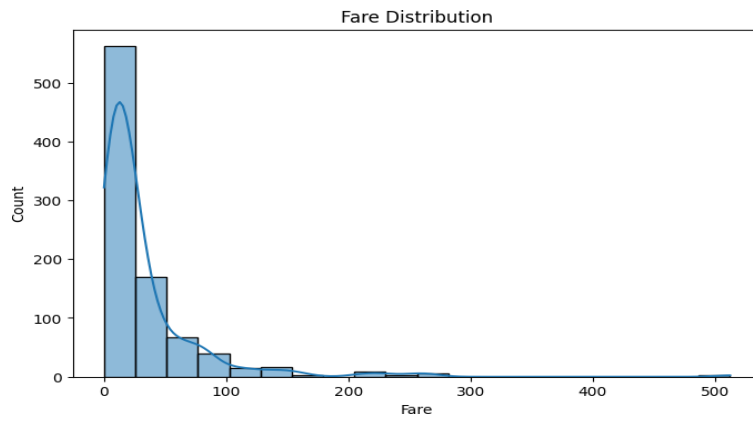
Missing Values:

PassengerId 0
Survived 0
Pclass 0
Name 0
Sex 0
Age 177
SibSp 0
Parch 0
Ticket 0
Fare 0
Cabin 687
Embarked 2
dtype: int64

Cleaned Dataset:

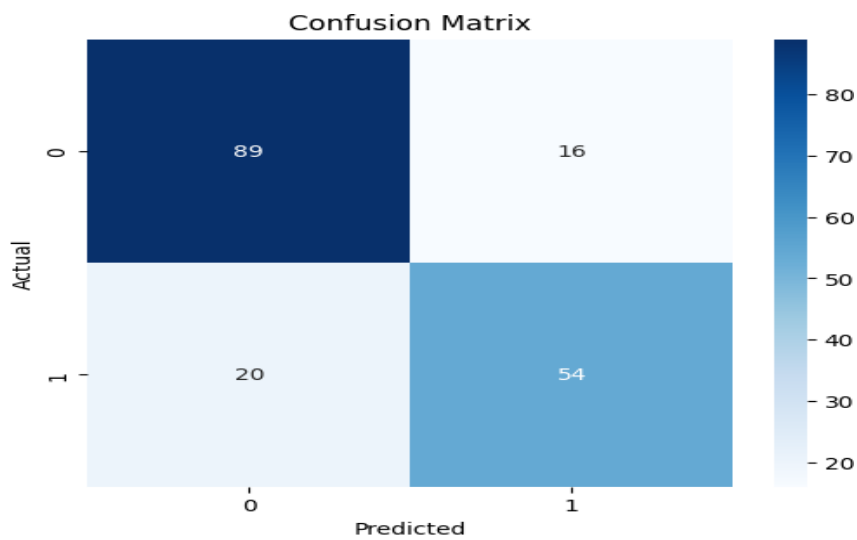
	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	0	22.0	1	0	7.2500	0
1	1	1	1	38.0	1	0	71.2833	1
2	1	3	1	26.0	0	0	7.9250	0
3	1	1	1	35.0	1	0	53.1000	0
4	0	3	0	35.0	0	0	8.0500	0





Model Accuracy:
79.89%

Classification Report:				
	precision	recall	f1-score	support
0	0.82	0.85	0.83	105
1	0.77	0.73	0.75	74
accuracy			0.80	179
macro avg	0.79	0.79	0.79	179
weighted avg	0.80	0.80	0.80	179



Model saved as titanic_model.pkl